

C-CLARKE INTERNATIONAL INSTITUTE OF DIGITAL SCIENCES

# Use Case Diagrams

Actors and use cases for the customer, provider, and admin workspaces.

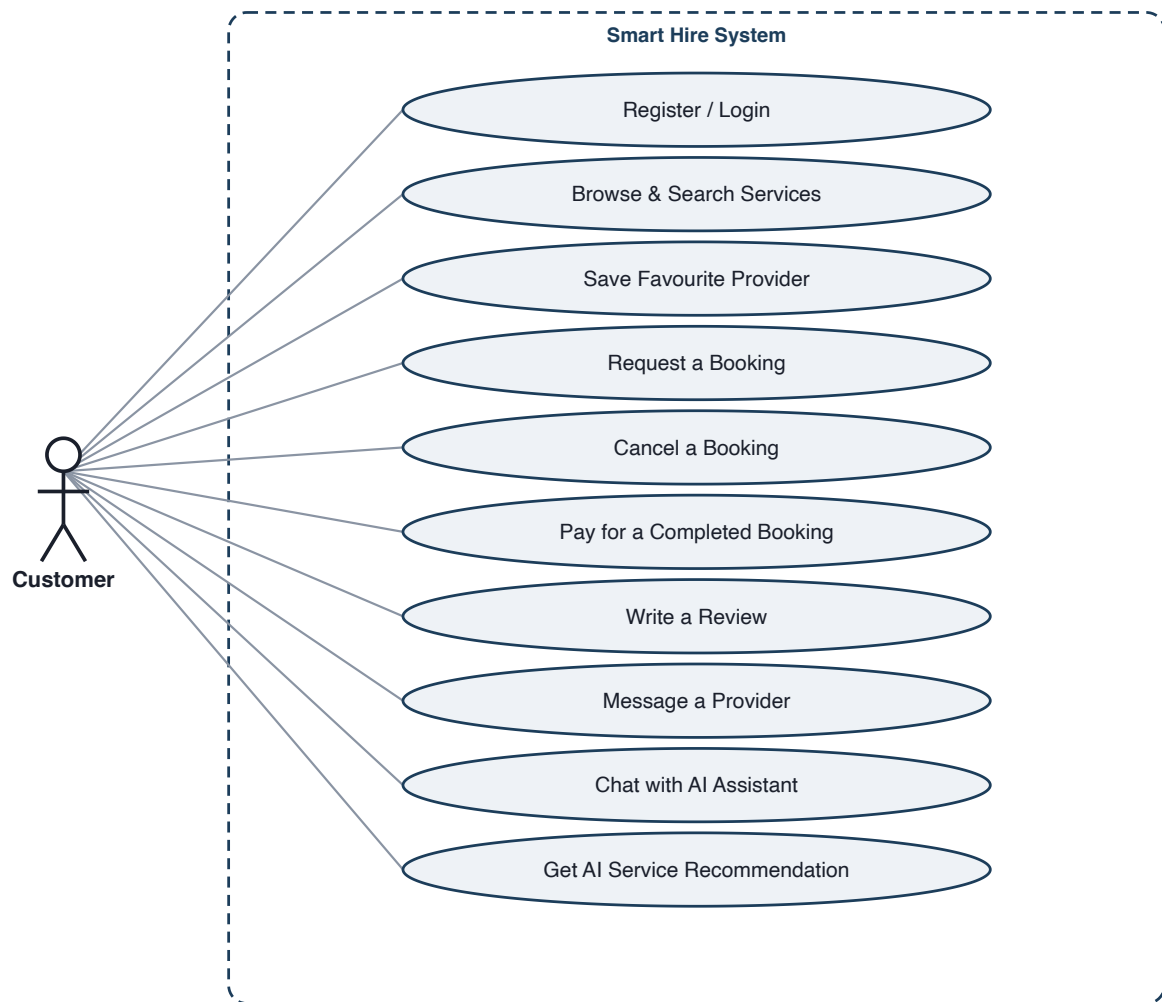
|                |                                                                               |
|----------------|-------------------------------------------------------------------------------|
| <b>Program</b> | Diploma in Advanced AI & Software Engineering                                 |
| <b>Batch</b>   | 25/26                                                                         |
| <b>Project</b> | Smart Hire — Service Marketplace                                              |
| <b>Team</b>    | Nilan Iddawela, Janaka Eranda, Dasun Anjana, Damith Lasantha, Matheesha Milan |
| <b>Date</b>    | 21 July 2026                                                                  |

# 1. Actors

| ACTOR                         | DESCRIPTION                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------|
| Customer                      | Registered user who searches for, books, pays for, reviews, and messages providers. |
| Service Provider              | Registered user who publishes services, manages availability, and fulfils bookings. |
| Administrator                 | Platform operator who moderates users, content, and payments.                       |
| OpenAI API (supporting actor) | External system that powers the AI assistant's replies and summaries.               |

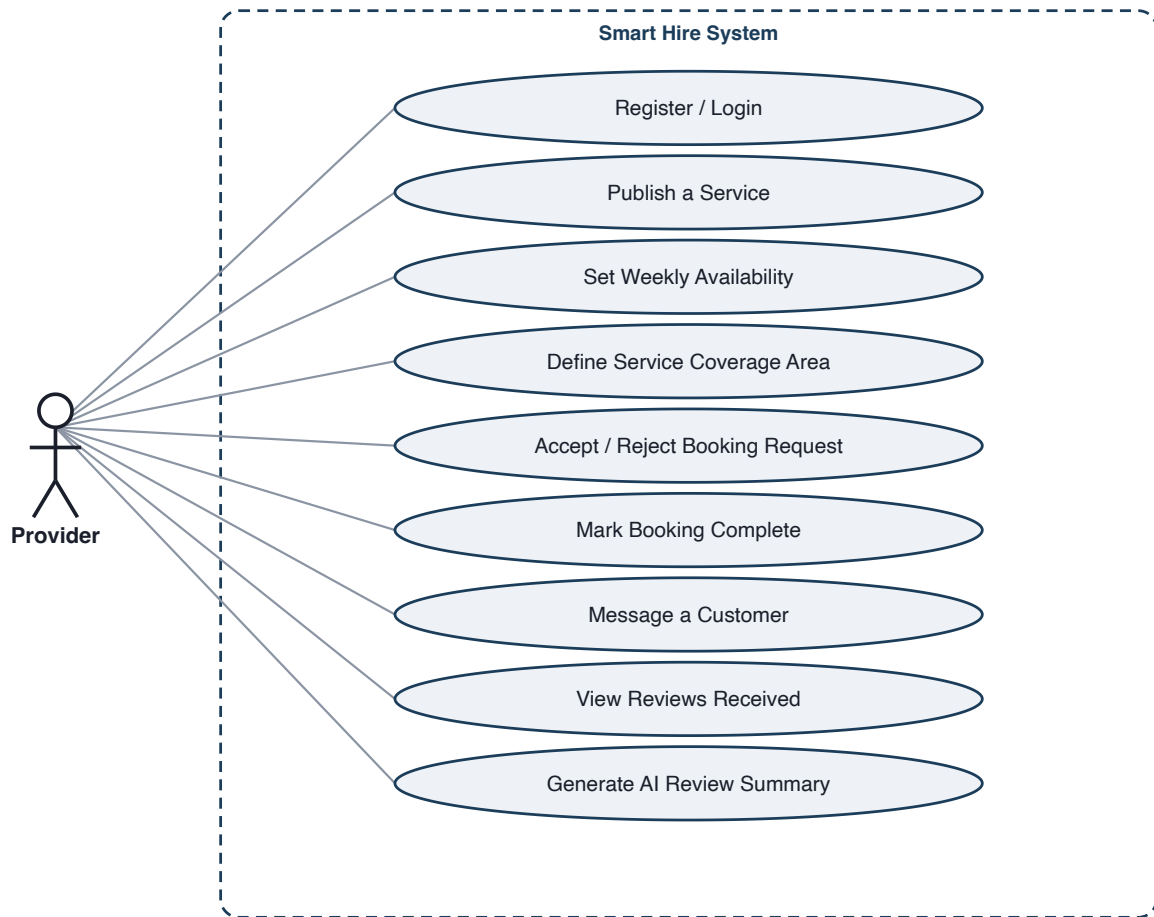
## 2. Customer Use Case Diagram

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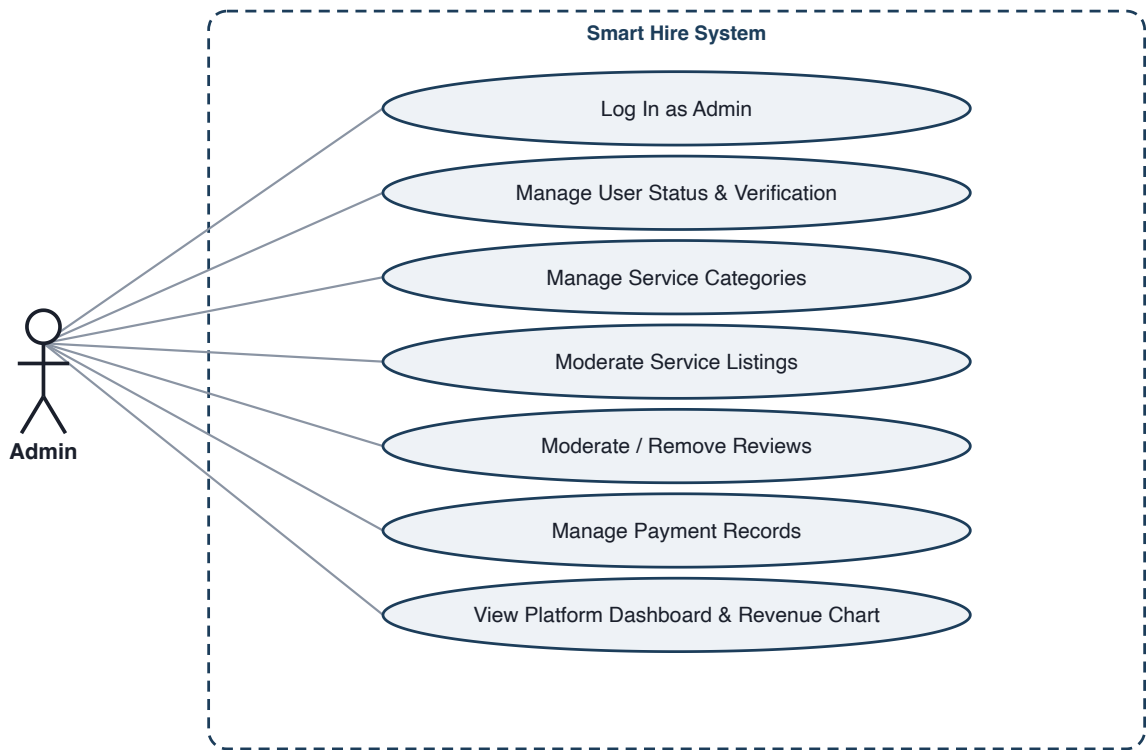


### 3. Service Provider Use Case Diagram

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## 4. Administrator Use Case Diagram



## 5. Detailed Use Case Descriptions

### UC-1: Request a Booking

|                |                                                                                                                                                                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor          | Customer                                                                                                                                                                                                                                                          |
| Precondition   | Customer is logged in; the service is active.                                                                                                                                                                                                                     |
| Main Flow      | <ol style="list-style-type: none"><li>Customer selects a service and a future date/time.</li><li>Customer submits the booking request with optional notes.</li><li>System creates the booking in <code>pending</code> status and notifies the provider.</li></ol> |
| Alternate Flow | Customer asks the AI assistant to book instead; the assistant checks real availability, confirms details in chat, then calls the same booking logic.                                                                                                              |
| Postcondition  | A real booking row exists; the provider sees it in their booking requests.                                                                                                                                                                                        |

## UC-2: Accept / Reject a Booking Request

|                      |                                                                                                                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>         | Service Provider                                                                                                                                                                                                                                     |
| <b>Precondition</b>  | A pending booking exists for one of the provider's services.                                                                                                                                                                                         |
| <b>Main Flow</b>     | <ol style="list-style-type: none"><li>1. Provider reviews the request.</li><li>2. Provider accepts (status → <code>accepted</code> ) or rejects (status → <code>rejected</code> ).</li><li>3. System notifies the customer of the outcome.</li></ol> |
| <b>Postcondition</b> | Booking status is updated; customer can message the provider once accepted.                                                                                                                                                                          |

## UC-3: Message a Provider

|                      |                                                                                                                                                                                                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>         | Customer, Service Provider                                                                                                                                                                                                                                              |
| <b>Precondition</b>  | A booking exists between the two parties.                                                                                                                                                                                                                               |
| <b>Main Flow</b>     | <ol style="list-style-type: none"><li>1. Either party opens the Messages page and selects the conversation for that booking.</li><li>2. They send a message.</li><li>3. The system stores it, notifies the other party, and shows an unread badge until read.</li></ol> |
| <b>Postcondition</b> | Message is stored against the booking; only that booking's customer and provider can read or send in it.                                                                                                                                                                |

## UC-4: Moderate a Review

|                      |                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>         | Administrator                                                                                                                                                                                                |
| <b>Precondition</b>  | Admin is logged in; a review exists.                                                                                                                                                                         |
| <b>Main Flow</b>     | <ol style="list-style-type: none"><li>1. Admin opens the review moderation page.</li><li>2. Admin inspects flagged or reported reviews.</li><li>3. Admin removes a review found to violate policy.</li></ol> |
| <b>Postcondition</b> | Review is removed from the system.                                                                                                                                                                           |